

- RAG Learning

Topic 8 — Energy: Forces Doing Work

Edexcel GCSE (9-1) Physics (1PH0) — Paper 2

8.1 Work Done

Work done by a force = **force × distance moved in the direction of the force**. It equals the energy transferred.

$$W = F \times d \quad (\text{joule, J; } 1 \text{ J} = 1 \text{ N}\cdot\text{m})$$

Worked Example: A 25 N force pushes a box 3 m horizontally in the direction of the force.
 $W = 25 \times 3 = \mathbf{75 \text{ J}}$. This 75 J is transferred to kinetic energy (and heat via friction).

Exam Tip: If the force is perpendicular to the motion, it does *no* work (e.g. gravity on a horizontal slide).

8.2 Power

$$P = E \div t = W \div t$$

Unit: watt (W) = 1 J/s. Also $P = F \times v$ (force × velocity) for a constant speed.

Worked Example: A motor lifts a 20 kg load 4 m in 5 s ($g = 10$).
Useful work = $mgh = 800 \text{ J}$. Useful power = $800/5 = \mathbf{160 \text{ W}}$.

8.3 Efficiency of Machines

$$\eta = \text{useful energy out} \div \text{total energy in} (\times 100\%)$$

In real machines, **friction** and **resistance** dissipate energy as heat and sound. Reduce by lubrication, streamlining, and good bearing design.

8.4 Energy and Machines

- A **lever** applies a smaller effort force over a larger distance to move a larger load over a smaller distance.
- A **pulley** can change the direction of a force and reduce the effort needed.
- A **gear** changes the trade-off between torque and rotational speed.

Key Points Summary

- Work done = $F \times d$ (in direction of F); unit joule.
- Power = W/t ; unit watt.
- Efficiency is always $< 100\%$ due to friction/heat/sound.
- Machines trade force for distance (or vice versa) but can't beat energy conservation.